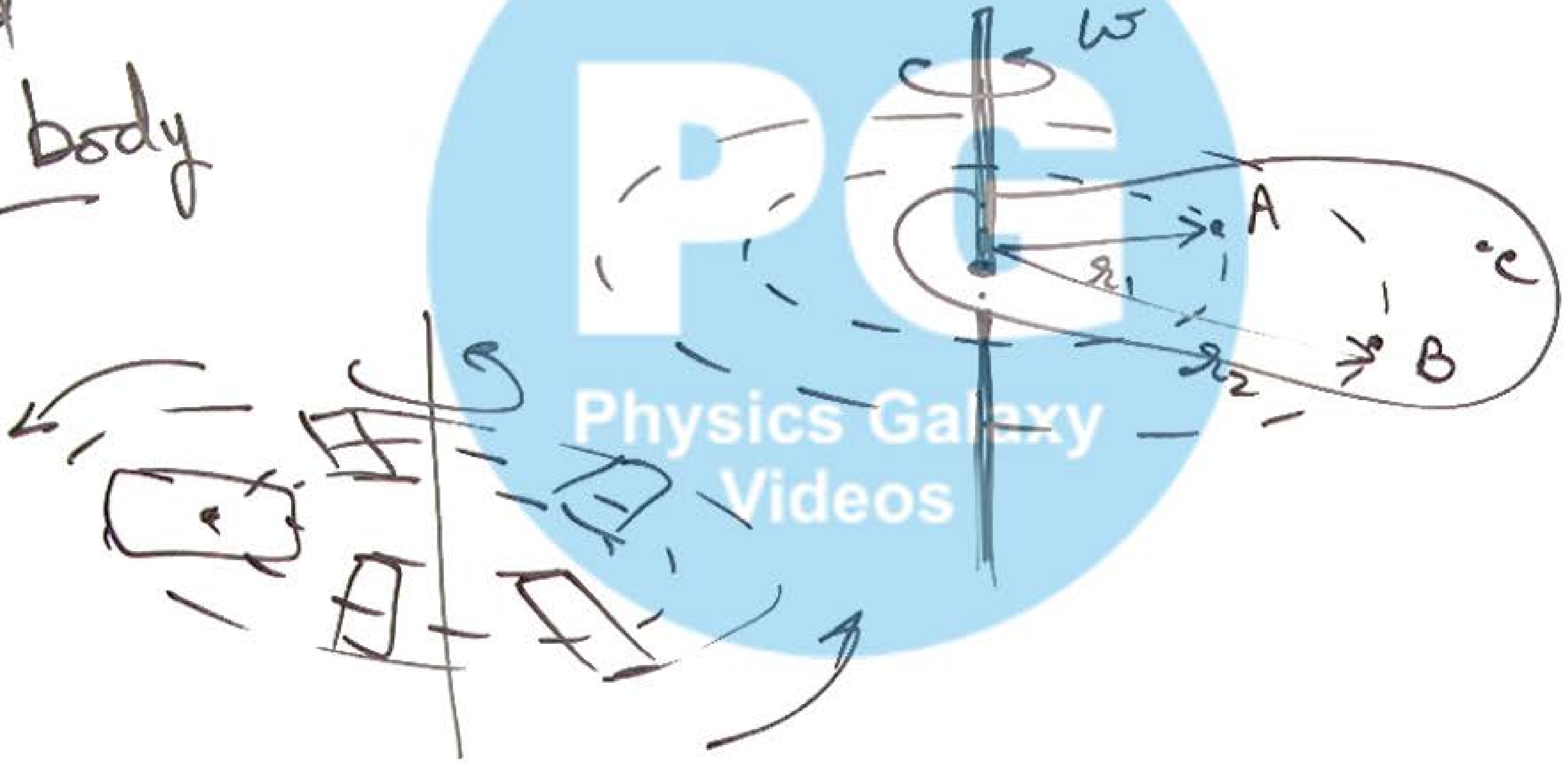
Rigid Body Dynamics-Part I

① Difference in Circular & Rotational Motion:

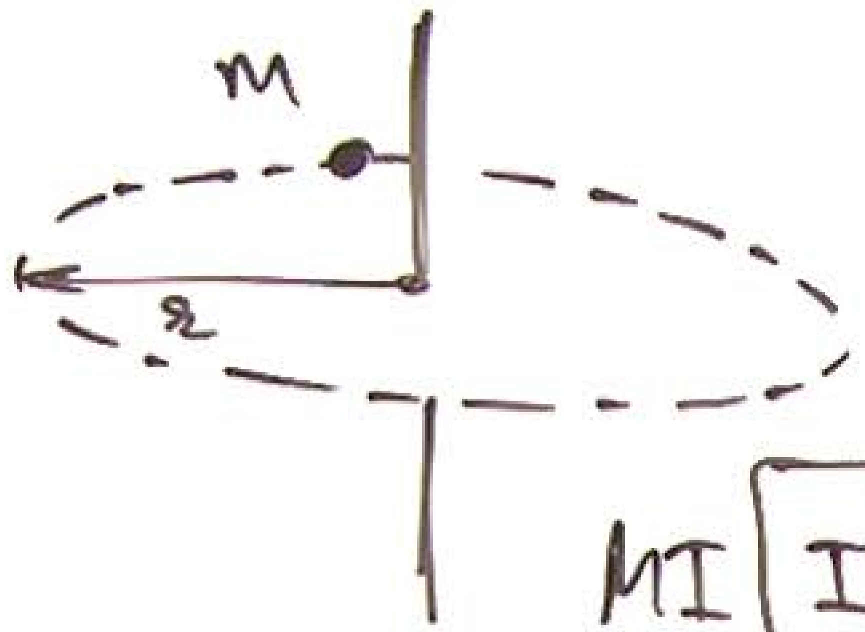
→ point mass

↓

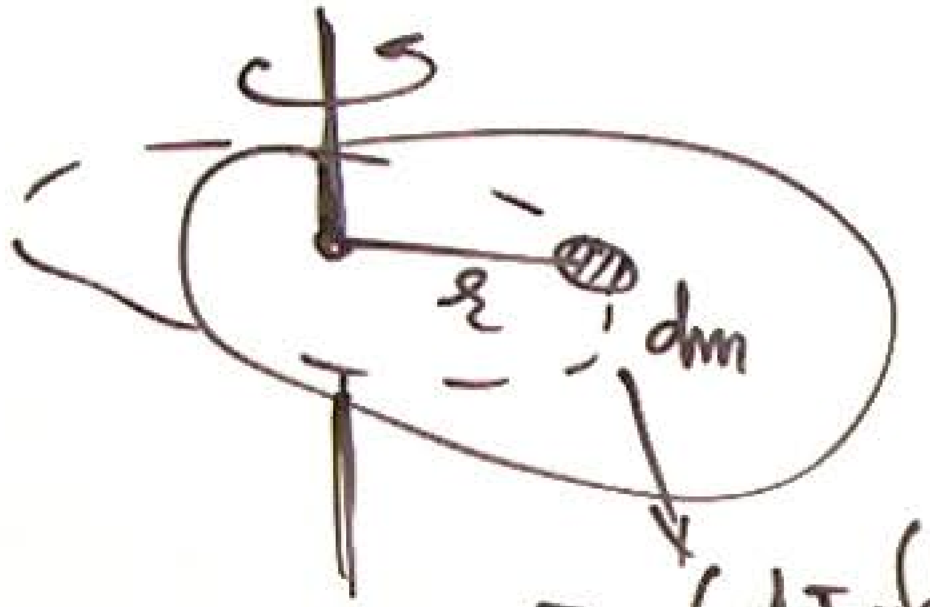
extended rigid body



2 Moment of Inertia:

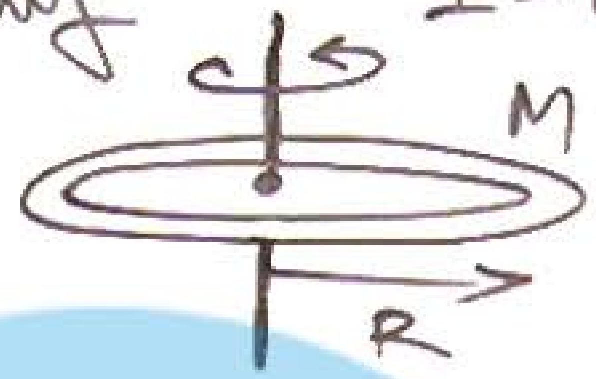


MI $\boxed{I = mr^2}$



$I = \int dI = \int dm\, r^2$

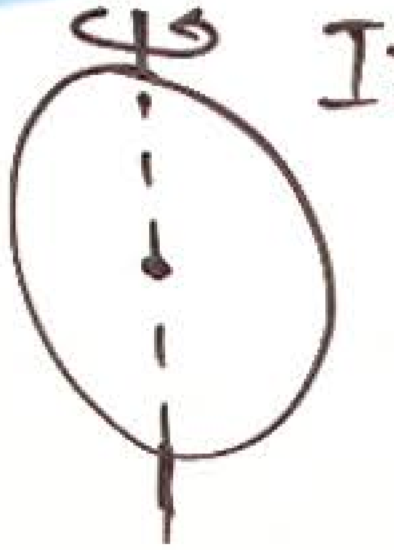
Ring $I = MR^2$



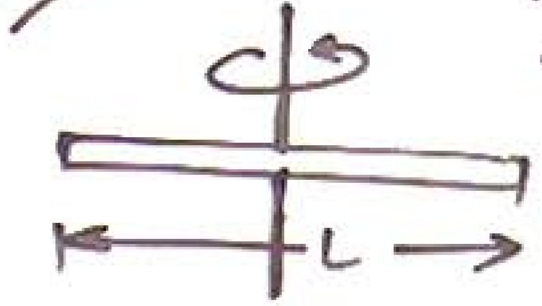
Disc $I = \frac{1}{2}MR^2$

Hollow Sphere $I = \frac{2}{3}MR^2$

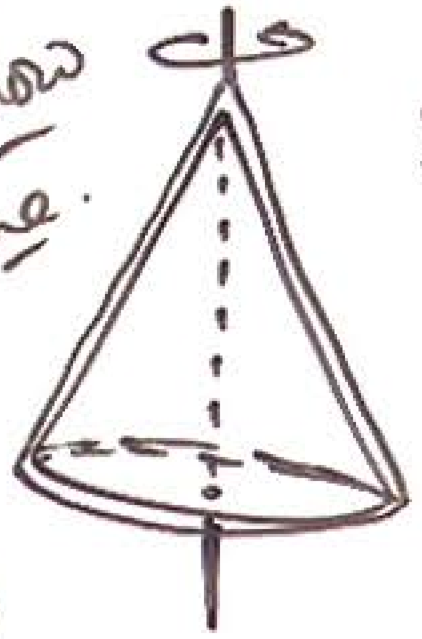
Solid Sphere $I = \frac{2}{5}MR^2$



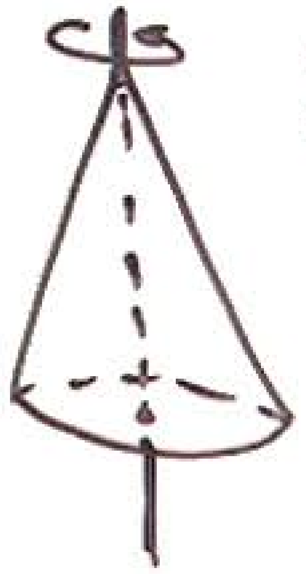
Thin rod $I = \frac{ML^2}{12}$



Hollow Cone $I = \frac{1}{2}MR^2$



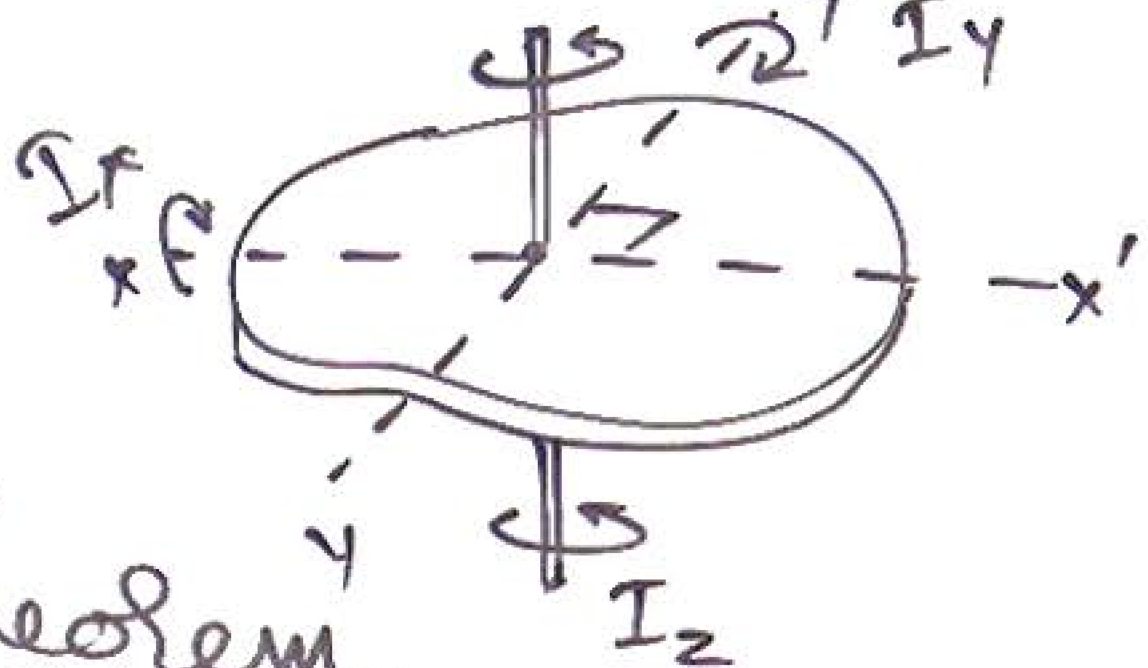
Solid Cone $I = \frac{3}{10}MR^2$



(3) Axes Theorems for MI :

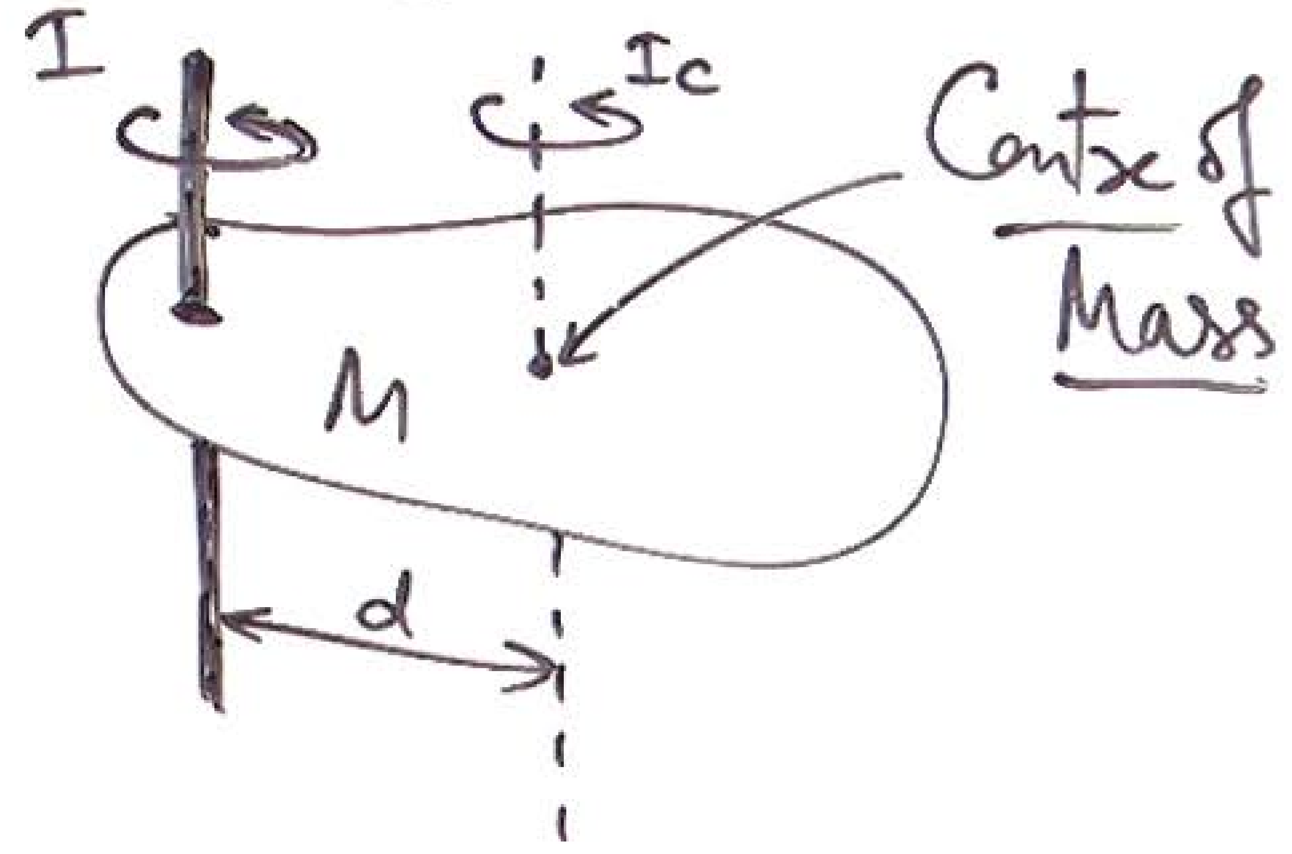
(1) Perpendicular Axes Theorem

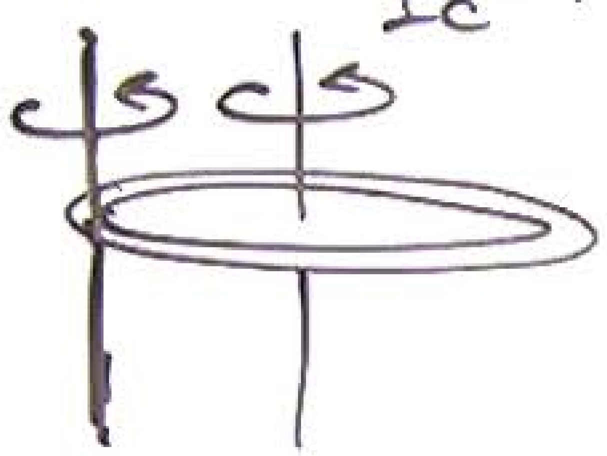
$$\boxed{I_z = I_x + I_y}$$



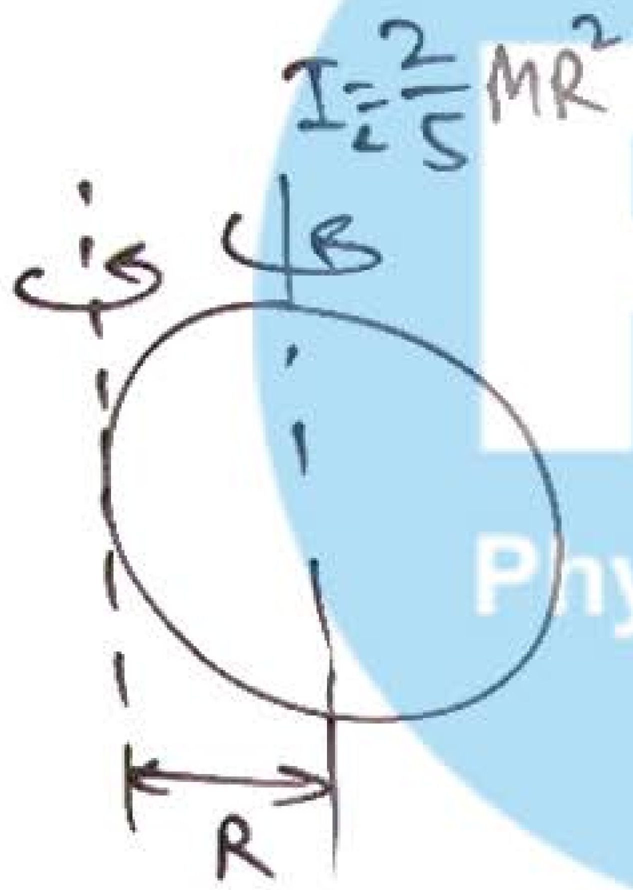
(2) Parallel Axes Theorem

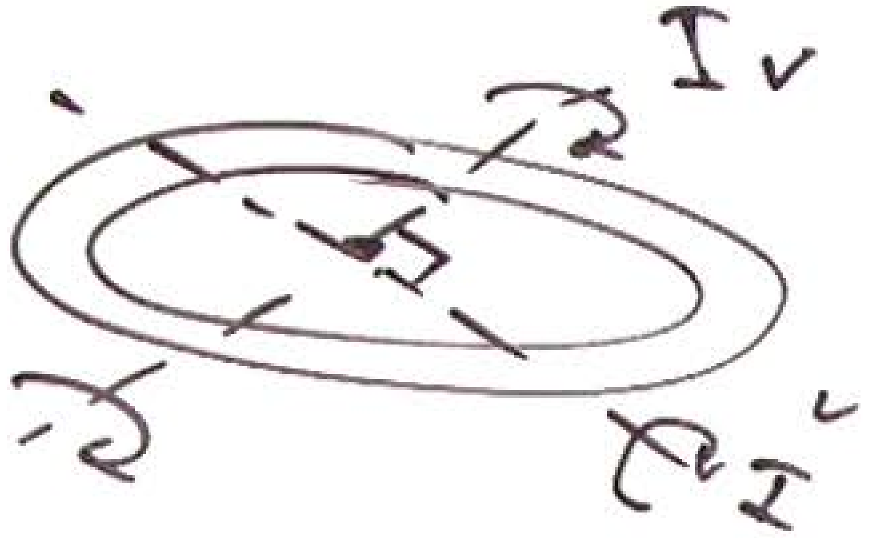
$$\boxed{I = I_c + Md^2}$$





$I = MR^2 + MR^2 = 2MR^2$





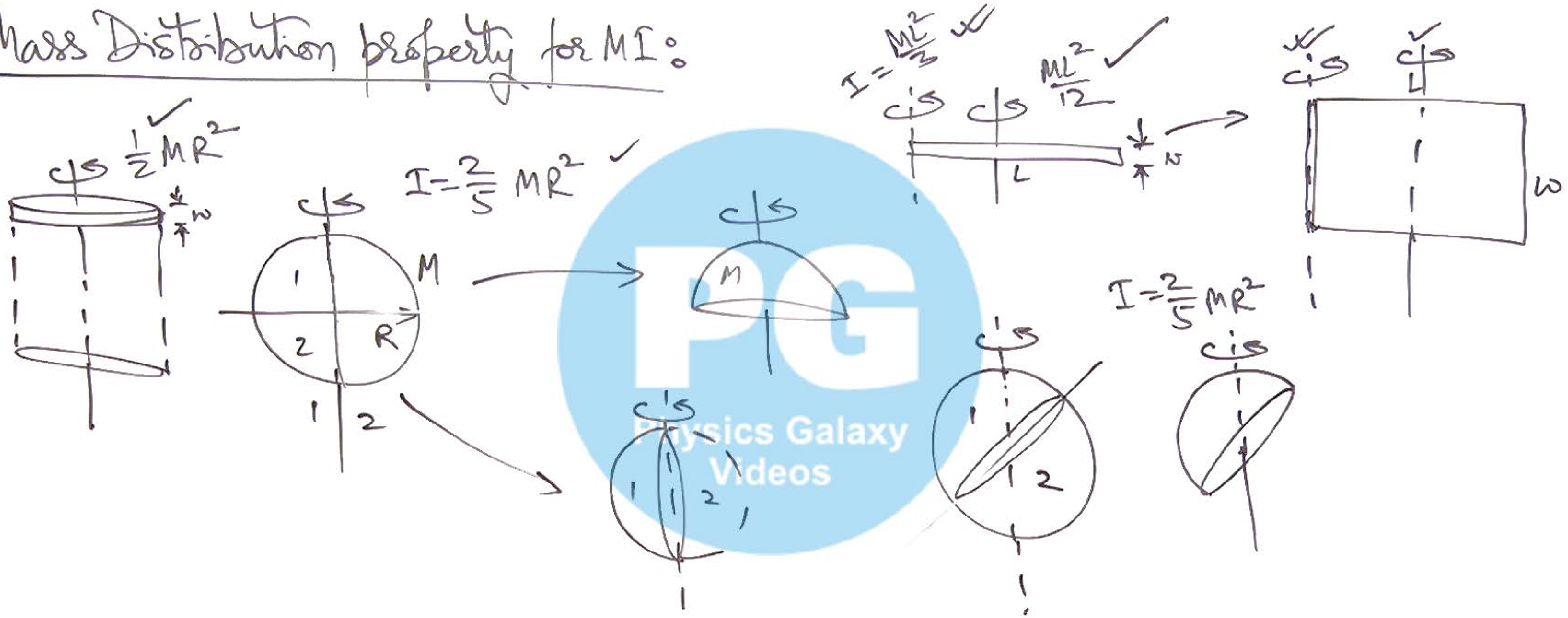
$I = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$

$2I = MR^2$

$$\boxed{I = \frac{1}{2}MR^2}$$

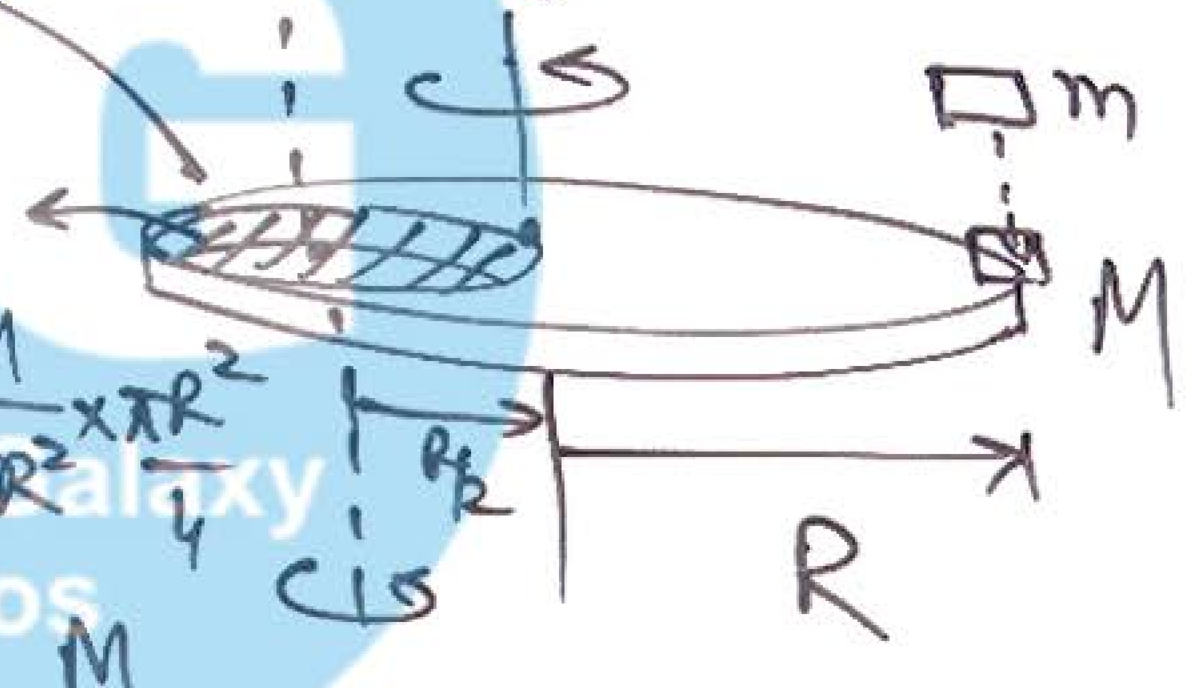
Rigid Body Dynamics-Part I

④ Mass Distribution property for MI :



(5) MI can be added or subtracted about given AoR:

$I_1 = \frac{1}{2}MR^2 \longrightarrow I_2 = \frac{1}{2}MR^2 + mR^2$



$m_1 = \frac{M}{\pi R^2} \times \frac{\pi R^2}{4}$

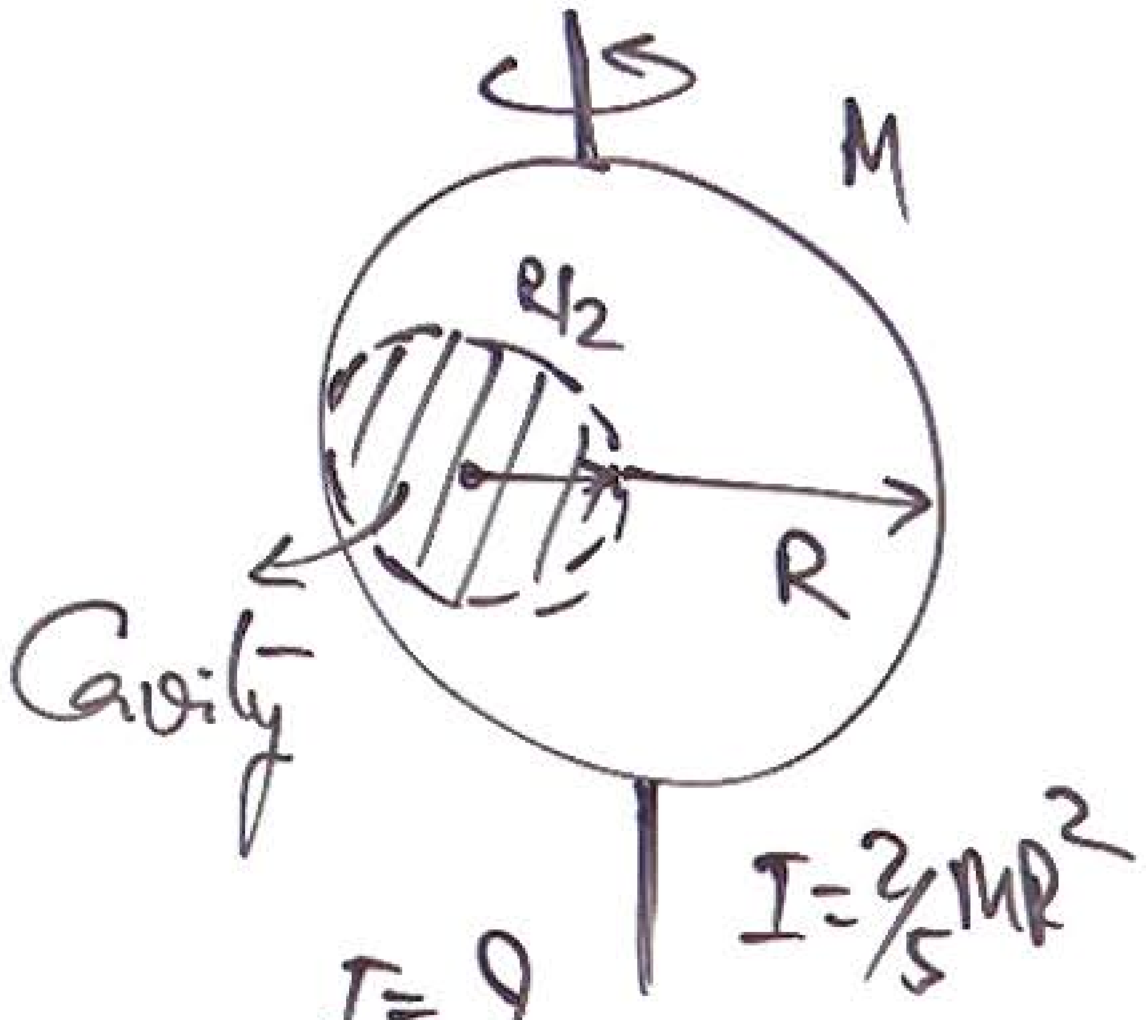
$m_1 = \frac{M}{4}$

$I_3 = \frac{1}{2} m_1 \frac{R^2}{4}$

$I_4 = I_2 - I_3$

$= \frac{1}{2}MR^2 + mR^2 - \frac{3}{2}\left(\frac{M}{4}\right)\left(\frac{R^2}{4}\right)$

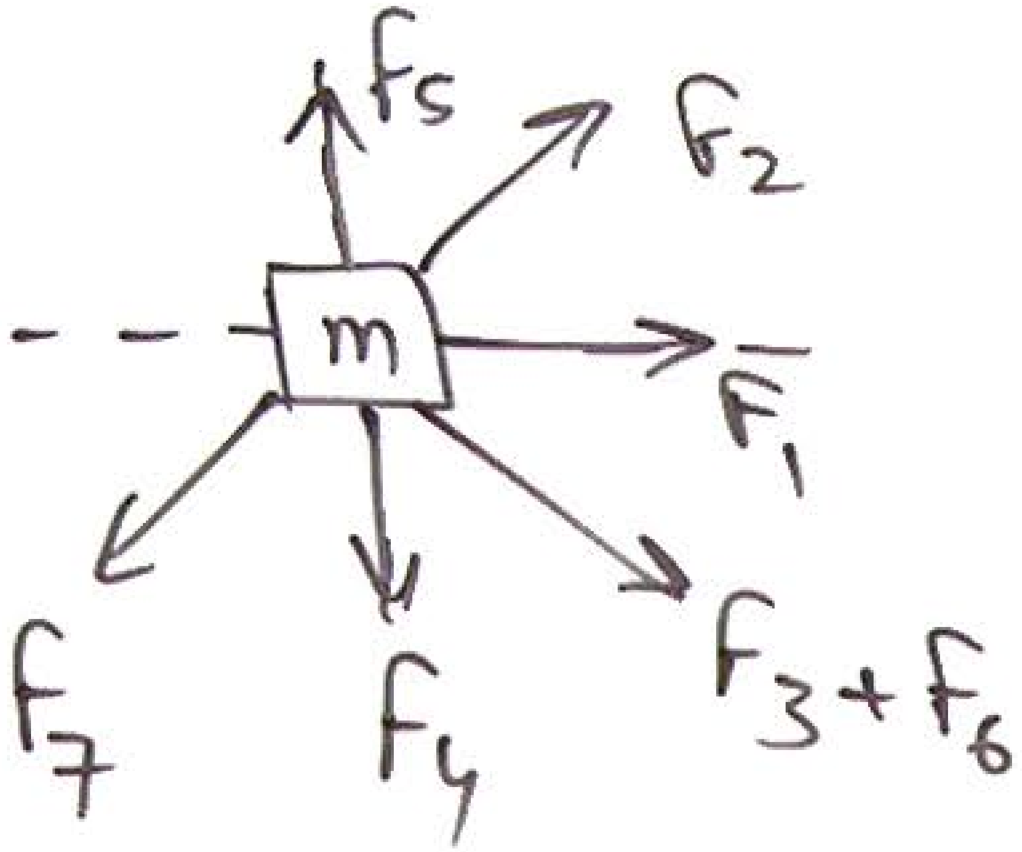
$= \_\_\_\_$



$I = \frac{2}{5}MR^2$

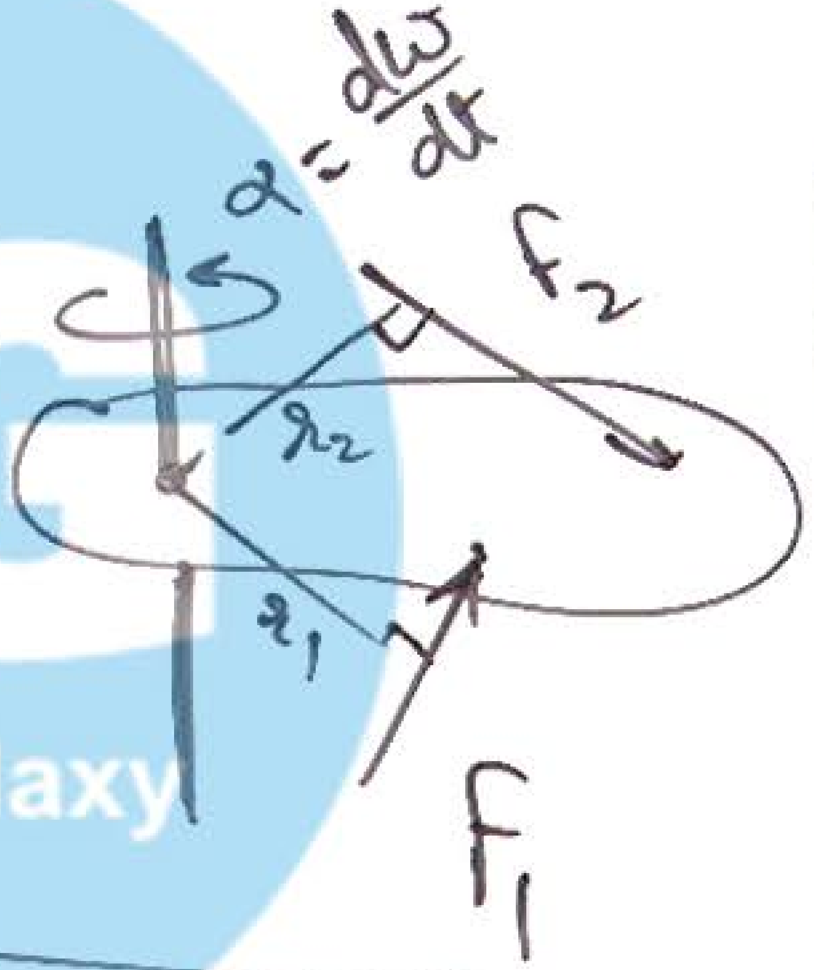
$I_{hem} = ?$

(6) Newton's II Law in Rotational Motion:



$$\Sigma F_x = m a_x$$

$$\Sigma F_y = m a_y$$



$$F_1 r_1 - F_2 r_2 = I \alpha$$

N II L.

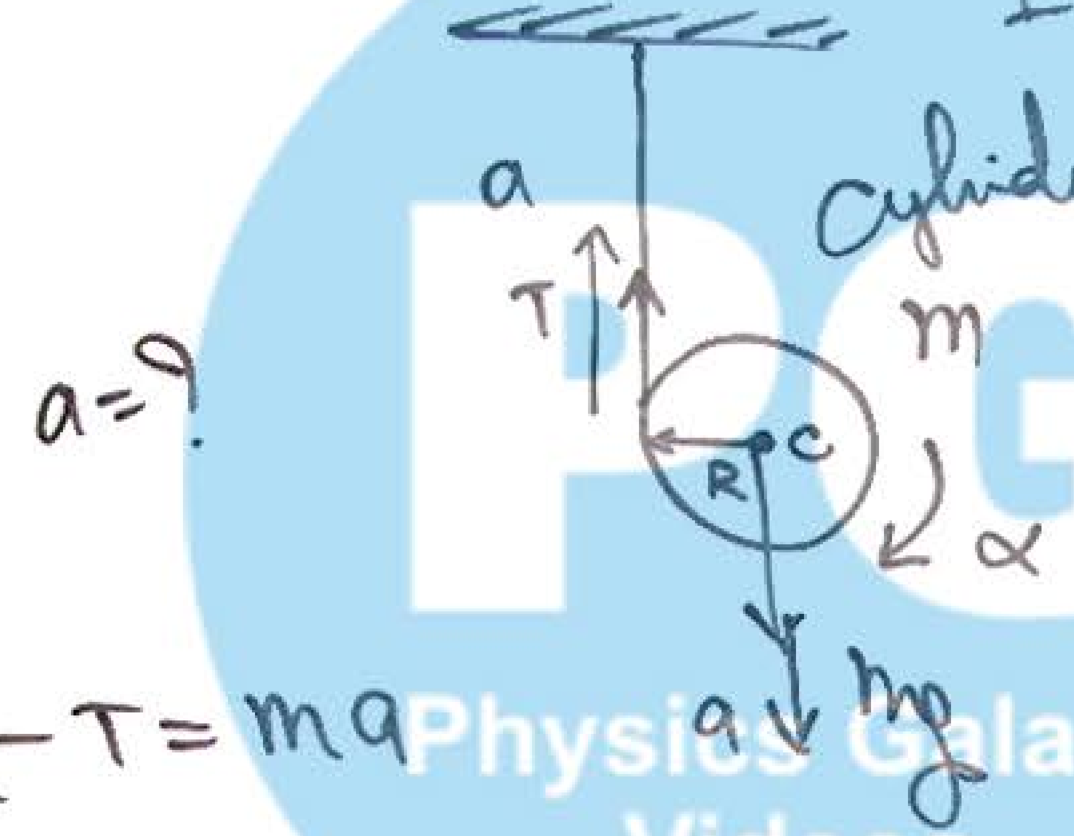
$$\boxed{\Sigma \vec{\tau} = I \vec{\alpha}}$$

(6) Newton's II Law in Rotational Motion :

$$\boxed{a_t = R\alpha.}$$

$I = \frac{1}{2}MR^2$

cylinder



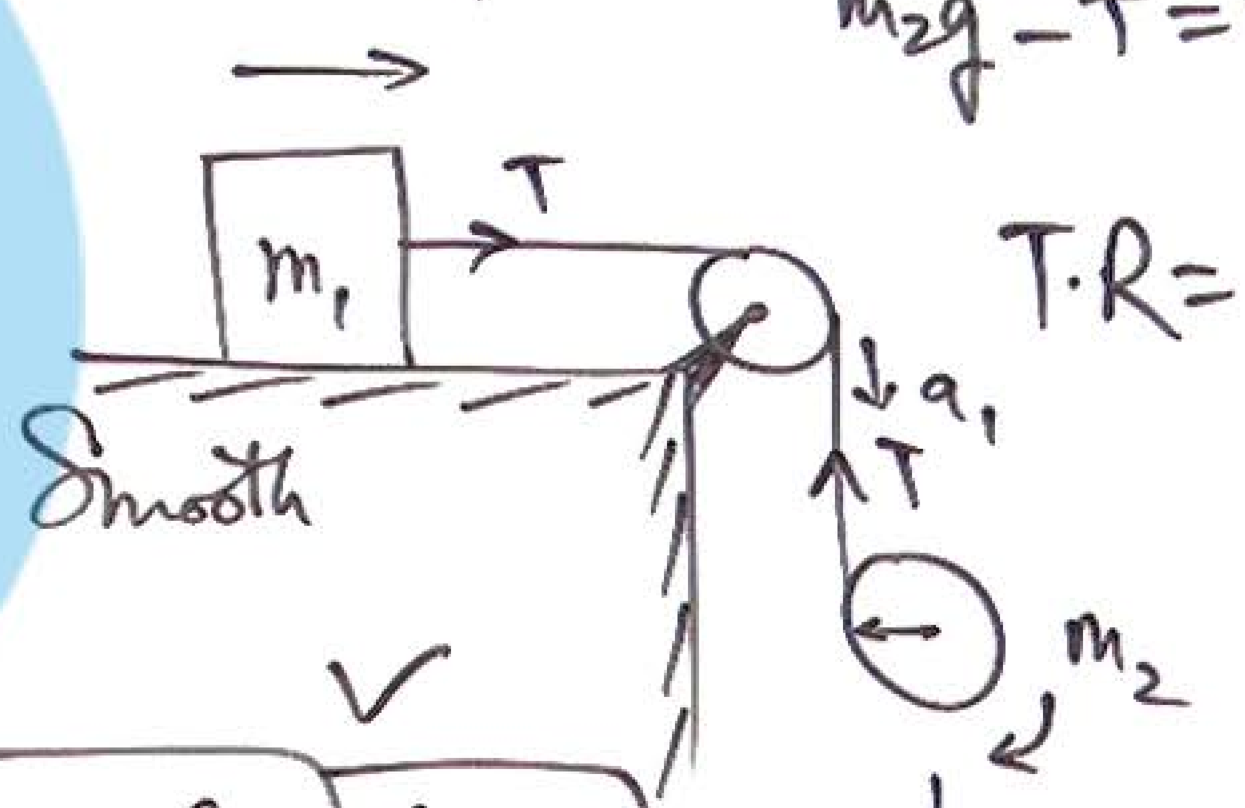
$mg - T = ma$

$mg = \frac{3}{2}mg$

$a = \frac{2}{3}g$ Ans

$TR = \left(\frac{1}{2}MR^2\right)\alpha$

$T = \frac{1}{2}mR \times a/R$



$$\boxed{\alpha = \frac{a_t}{R} = \frac{a_2 - a_1}{R}}$$

$T = ma_1$

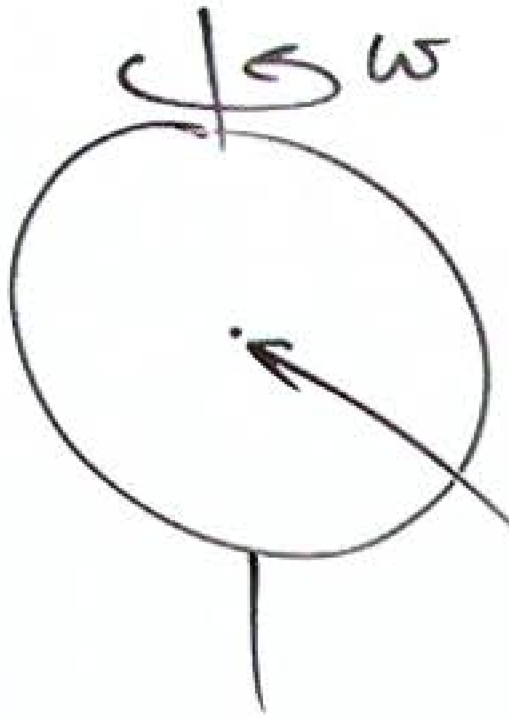
$m_2 g - T = m_2 a_2$

$T \cdot R = I\alpha$

$a_2 = a_1 + R\alpha$

(7) Concept of "Static Equilibrium":

$$\left[\begin{array}{c}\vec{V}_{cm}=0\\ \&\\ \vec{\omega}=0\end{array}\right]$$



COM is at rest

$$\sum \vec{F}_{net}=0 \Rightarrow \vec{a}_{CM}=0 \rightarrow v_{cm}=0 \rightarrow \text{Static eq}^{m} \text{ of COM of body}$$

$$\vec{a}_{CM}=0 \rightarrow V_{cm}=\text{Const}$$

Static eq^{m} $\rightarrow$ $\sum \vec{F}_{net}=0$ & $\sum \vec{\tau}_{net}=0$

$$\left(\begin{array}{c}V_{cm}=0\\ \\ \omega_{body}=0\end{array}\right)$$